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Live Class – Diwali Special

Topic : Quadratic Equations - New TCS Type

Type 1:Concept Sheet

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Question 1 :

If $\propto$ *and* β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\propto} + \frac{1}{\beta}$ is:

a. $\frac{4}{7}$ b. $\frac{-3}{7}$ c. $\frac{3}{7}$ d. $\frac{-3}{4}$

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Question 2:

If $\propto$ *and* β are the roots of the equation $x^2 - x + 1 = 0$, then write the value $\propto^2 + \beta^2$

a. 1 b. –1 c. 0 d. None of these

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Question 3:

If the equation $2x^2 - 7x + 12 = 0$ has root $\propto$ *and* β , then the value of $\frac{\propto}{\beta} + \frac{\beta}{\propto}$

a. $\frac{97}{24}$ b. $\frac{7}{2}$ c. $\frac{1}{24}$ d. $\frac{7}{24}$

Type 2:Concept Sheet

Question 4:

One root of quadratic equation x^2 - kx + 27 = 0 is 3, then find the value of 'k'

a. 10 b. 12 c. –12 d. 16

Question 5:

Find the value of k if one root of the equation: $x^2 - 9x + k = 0$ is twice the other root.

a. 18 b. 16 c. 12 d. 9

Question 6:

Find the value of k so that the sum of the roots of the equation

$3x^2 + (2k + 1)x - k - 5 = 0$ is equal to the product of the roots.

Type 3:Concept Sheet

Question 7(a):

If the sum of the roots of a quadratic equation is 1 and product of the roots is –20. Find the quadratic equations.

a. $x^2 - x - 20 = 0$
b. $x^2 + x + 20 = 0$
c. $x^2 + x - 20 = 0$
d. $x^2 - x + 20 = 0$

Question 7(b):

Which of the following quadratic equation has roots –3 and –5

a. $x^2 - 8x + 15 = 0$
b. $x^2 - 8x - 15 = 0$
c. $x^2 + 8x + 15 = 0$
d. $x^2 + 8x - 15 = 0$

Type 4:Concept Sheet

If α and β are the roots of equation :

$$x^2 - x + 1 = 0$$

then which equation will have roots α^3 and β^3

1. $x^2 - 2x + 1 = 0$
2. $x^2 + 2x + 1 = 0$
3. $4x^2 - 3x + 2 = 0$
4. $3x^2 - 5x + 2 = 0$

Q9. If $\propto, \beta$ are roots of the equations $x^2 - 5x + 6 = 0$, then find the quadratic equation whose roots are $\frac{1}{\propto}, \frac{1}{\beta}$

a. $6x^2 - 5x + 1 = 0$

b. $6x^2 + 5x + 1 = 0$

c. $6x^2 - 5x - 1 = 0$

d. $6x^2 + 5x - 1 = 0$

If α and β are the roots of equation :
$x^2 - 3x + 2 = 0$

then quadratic equation whose have roots are (α +1) and (β +1) is ?

1. $x^2 - 5x + 6 = 0$
2. $x^2 + 5x + 6 = 0$
3. $x^2 - 5x + 2 = 0$
4. $x^2 - 5x - 6 = 0$

Type 5:Concept Sheet

Type:

A quadratic eqution is of the form $ax^2 + bx + c = 0$, where x is an unknown variable, a,b,c are constants and $a \neq 0$

$ax^2 + bx + c = 0$ is the standard form of quadratic equation.

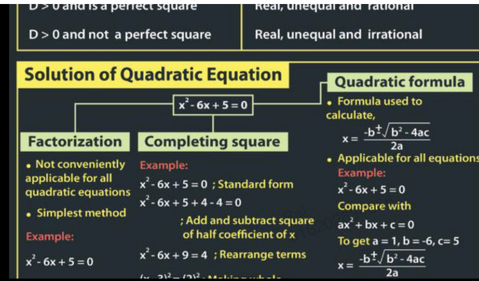
Root of Quadratic Equation

It is the value of the unknown variable for which the quadratic equation holds true. A quadratic equation has two roots.

Nature of Roots

For equation $ax^2 + bx + c = 0$, expression $b^2 - 4ac$ is called Discriminant and denoted by D.

Value of Discriminant	Nature of roots
$D < 0$	Unequal and imaginary
$D = 0$	Real and equal



Question 1: Which of the following quadratic Equations has real roots?

1. $5x^2 - 2x + 2 = 0$
2. $3x^2 - 4x + 2 = 0$
3. $4x^2 - 3x + 2 = 0$
4. $3x^2 - 5x + 2 = 0$

Question 2 :

For what positive values of m , does the equation

$(3m+1)\,x^2+2(m+1)\,x+m=0$

have real and equal roots ?

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